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Application Credentials
Ziyi Chen
Feb 6, 2026

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Recommendation from Prof. Heng Huang	Confidential Letter of Recommendation or Evaluation	CONFIDENTIAL
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November 10, 2024

Dear Faculty Search Committee,

It is my great pleasure to write this strong letter of recommendation for Ziyi Chen for a tenure-track faculty position.

Ziyi is my postdoctoral researcher in the Computer Science Department at the University of Maryland, College Park. He is a great researcher, collaborator and teacher, as will be elaborated next.

Ziyi is a productive researcher. He has led and assisted with many research projects on machine learning algorithms with theoretical foundation, which have been respectively published on ICLR 2024, ICML 2024, Neurips 2024, and submitted to ICLR 2025 under review. During this research process, he shows his independence, persistence, creativity, deep insights and strong math skills. Below, I will briefly overview some of his important works in my lab, where he led the whole research process including lit search, algorithm design, theoretical derivation, experiments and writing.

Constrained Cooperative Reinforcement Learning (ICLR 2024)

Ziyi's first work at my lab is a comprehensive theoretical study on constrained cooperative reinforcement learning, an emerging area aiming to train multiple intelligent agents to complete a collaborative task in a dynamic environment with safety requirements. In this work, Ziyi for the first time proves that the mainstream approach to constrained cooperative MARL, namely, the primal-dual algorithm, may fail to converge to the global optimal solution due to nonzero duality gap. Instead, he derives a new global convergence rate of the primal-dual algorithm that contains this nonzero duality gap. Then he designs a fully decentralized primal algorithm and derives its global convergence rate that contains an alternative constant gap. Interestingly, he proposes numerical examples which demonstrate that none of the two algorithms can outperform the other. This work has been accepted by the ICLR 2024 conference.

Accelerated Robust Reinforcement Learning (ICML 2024)

Robust reinforcement learning is an important machine learning problem that aims to train the agents to make the optimal decision that is robust to environmental change. For the popular and tractable class of s-rectangular robust reinforcement learning, Ziyi proposes an accelerated policy gradient algorithm which improves the state-of-the-art theoretical complexity. Furthermore, Ziyi extends the convergence results from fully known environment (deterministic setting) and small state spaces used in existing works, to the practically much more widely used unknown environment (stochastic case) and large state spaces. This work has been accepted by the ICML 2024 conference.

Robust Reinforcement Learning with General Utility (NeurIPS 2024)

In this work, Ziyi proposes a new reinforcement problem called robust reinforcement learning with general utility, which for the first time extends robust reinforcement learning from standard linear

utility to various general utilities (purposes), including exploitation (standard linear utility), exploration, exploitation-exploration trade-off, safe robust exploitation, learning from expert demonstrations, etc. Then he proposes two algorithms for his proposed new learning framework and derives their theoretical complexity results required to respectively converge to a stationary point and a global optimal solution. This work has been accepted by the NeurIPS 2024 conference.

Large Language Model Alignment (Under Review of ICLR 2025)

As large language models (LLM) like ChatGPT become quite popular, it is essential to align LLM with human values to make LLM helpful, honest, and harmless. The most commonly used alignment techniques are reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO). Although RLHF and DPO have many successes, there are still obstacles to achieve effective alignment. First, human preference data can be corrupted by personal bias, malicious falsification, etc. Second, optimization over the inaccurately estimated reward can produce responses with low actual quality (true reward). Third, LLM often favors too verbose but useless responses. Most existing works target only one of the above three issues (corruption, overoptimization, verbosity). Some other works need large amount of computation to estimate multiple reward models to solve these issues and have no theoretical foundation of generalization ability. To solve these three issues simultaneously, Ziyi proposes a new RLHF algorithm and its simplified equivalent DPO algorithm, supported by both generalization theory and experimental results. Moreover, his proposed DPO algorithm is almost as easy to implement as the vanilla DPO. This work is under review of ICLR 2025.

In the above works, Ziyi shows remarkable independence, persistence, creativity, deep insights and strong math skills. To overcome technical difficulties, he kept improving his obtained results until he felt satisfied. In particular, after finishing the first draft of the NeurIPS 2024 work, he found an additional novel algorithm and its global convergence result would greatly improve his work. Then he designed this algorithm and proved related theorems in just one week. Later, in the first two rounds of NeurIPS 2024 discussion, a reviewer gave only 3 scores (reject) with highly negative comments. In the first round, Ziyi clearly and politely presented his detailed plan to solve the reviewers' proposed issues. In the second round, this reviewer gave comments to Ziyi's four responses, and Ziyi accurately found that only two of them need to be addressed. To our surprise, after Ziyi briefly addressed these two questions, this reviewer improved the rating from 3 (reject) to 7 (accept), and this work was successfully accepted by NeurIPS 2024. Such a dramatic rating increase is rarely seen among the works of my students and postdocs.

Ziyi is not only a great researcher, but also a great collaborator. He has mentored and collaborated with five of my students and postdoctoral researchers, where he has provided thoughtful insights in improving idea novelty, proving theorems, designing and debugging experiments, etc. He is also good at using collaboration to help him quickly solve difficult problems in his own research work. For example, for his NeurIPS 2024, he proposed a theoretical hypothesis. He believed it is correct based on coding check and his strong intuition, but he did not find out how to prove this hypothesis. To accelerate the research, Ziyi simplified the difficult part of the problem into a math exercise. Then with my permission, he posted this exercise to my lab group, promising that anyone solving it would

be his co-author. Very soon, my exceptional PhD student Zhengmian Hu solved this problem and became the co-author. Conversely, Ziyi also largely helped and co-authored Zhengmian's work by proving a key theorem which reduces the problem dimensionality from exponential to linear and thus makes the algorithm tractable.

Ziyi is also a great teacher. He has co-taught two lectures on reinforcement learning for my machine learning course. His slides and presentation are clear with pictures of specific examples. For example, he explained the concept of reinforcement learning by an interesting example of a robot beginning to avoid fire after being burned. Then he explained key elements of reinforcement learning, including state, action and reward, in specific application scenarios including medical treatment, autonomous driving and board game.

In summary, Ziyi is an outstanding researcher, collaborator and teacher. I most enthusiastically recommend Ziyi to be an assistant professor in your department.

Sincerely,



Heng Huang
Brendan Iribe Endowed Professor
Department of Computer Science
Department of Electrical and Computer Engineering
University of Maryland Institute for Advanced Computer Studies
University of Maryland Institute for Health Computing
University of Maryland College Park

PS: I am a Brendan Iribe Endowed Professor in Computer Science at the University of Maryland College Park. I received the PhD degree in Computer Science at Dartmouth College. My research areas include machine learning, data mining, bioinformatics, and precision medicine. I have published more than 300 papers in top-tier conferences and many papers in premium journals, such as ICML, NeurIPS/NIPS, KDD, RECOMB, ISMB, ICLR, ICCV, CVPR, IJCAI, AAAI, Nature Machine Intelligence, Nature Communications, Nucleic Acids Research, Bioinformatics, Medical Image Analysis, IEEE TPAMI, TMI, TIP, TKDE, JMLR, etc. I am currently leading multiple NIH R01, U01, and NSF funded projects on machine learning, neuroimaging, precision medicine, electronic medical record, and IoT. I have received more than \$52M research funding from both NSF and NIH. I served as the Program Chair of ACM SIGKDD 2020. I am a Fellow of American Institute for Medical and Biological Engineering (AIMBE).